

The Matrix-Element Principle and Its Reach

What follows from $|\langle \nu_0 | R_5 | \nu_1 \rangle|^2 = 2/9$ beyond the lepton masses

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The Matrix-Element Principle and Its Reach

Abstract. The lepton masses follow without free parameters from a single matrix element, $|\langle v_0 | R_5 | v_1 \rangle|^2 = 2/9$. This document examines what the same mechanism carries beyond that. First result **[B]**: in the $\mathbb{Z}_3$ mode basis the icosahedral group A_5 generates exactly eight squared matrix-element moduli, $\{0, (5-3\phi)/9, 1/9, 2/9, 4/9, 5/9, (2+3\phi)/9, 1\}$; the reach of the mechanism is therefore exactly bounded. Second result **[K]**: the on-shell Weinberg angle $1 - M_W^2/M_Z^2$ lies within 0.4 % of $2/9$ — the on-shell scheme is the natural one for pole masses from $\tilde{T} \cdot m = 1$; the $\overline{\text{MS}}$ deviation of 3.9 % is scheme dependence. Prediction $M_W = 80.420$ GeV. Third result **[K]**: the atmospheric mixing angle satisfies $\sin^2 \theta_{23} \in \{4/9, 5/9\}$; the octant ambiguity of the data is the ambiguity $4/9 \leftrightarrow 5/9$, maximal mixing $1/2$ is excluded. Negative **[X]**: $\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$ and α_s do not lie in the eight-element set and cannot be reached by a single A_5 matrix element.

Notation

Epistemic markers:

- [B]** algebraically proven (check script, exact arithmetic)
- [K]** numerically confirmed
- [S]** structurally plausible, identification open
- [E]** established external mathematics or physics
- [X]** tested and negative

Symbol	Meaning
R_5	five-fold rotation (72°), order-5 element of A_5
v_0, v_1, v_2	orthonormal $\mathbb{Z}_3$ modes (symmetric, phase-carrying)
$p_k = \langle v_0 R_5 v_k \rangle ^2$	icosahedral redistribution weights
$\phi = (1 + \sqrt{5})/2$	golden ratio
$\theta = 2/9 = p_0$	Koide phase (= Born probability, Doc. 367)
$\xi = 4/30000$	FFGFT scaling parameter
$\sin^2 \theta_W$	Weinberg angle (electroweak mixing)
θ_C	Cabibbo angle (quark generation mixing)
C_3	three-fold rotation (120°), second generator of A_5
A_5	icosahedral group (60 elements)
$\mathbb{Z}_3$	cyclic group of order 3 (mode symmetry)
$\text{GF}(q), \text{GF}(q)^*$	Galois field with q elements, its multiplicative group
$\tilde{T}$	proper-time scaling ($\tilde{T} \cdot m = 1$)
Q	Koide quotient $\sum m_i / (\sum \sqrt{m_i})^2$
$Q_{\text{up}}, Q_{\text{down}}$	Koide quotient of the up and down quarks
r_i, p_i	prefactor and exponent of the mass formula $m_i = r_i \xi^{p_i v}$
K_{frak}	fractal correction factor $1 - 100\xi$ (Doc. 133)
M_W, M_Z	masses of the W and Z bosons (pole masses)
m_h, m_t, m_s	mass of the Higgs particle, of the top quark, of the strange quark
m_e, m_p	mass of the electron, of the proton
v	electroweak vacuum scale (246.22 GeV, SM definition)
$\overline{\text{MS}}$	renormalisation scheme (minimal subtraction)
$\theta_{12}, \theta_{13}, \theta_{23}$	neutrino mixing angles (solar, reactor, atmospheric)
δ_{CP}	CP-violating phase
$\Delta m_{\text{atm}}^2, \Delta m_{\text{sol}}^2$	differences of the squared neutrino masses
α, α_s	fine-structure constant, strong coupling
Λ_{QCD}	confinement scale of the strong interaction
R_{Proton}	charge radius of the proton
r_e, λ_p	classical electron radius, Compton wavelength of the proton
σ	standard deviation of the measurement

1 The Mechanism and Its Yield for the Leptons

The starting point of the mechanism is a single number: the squared modulus of the matrix element of the five-fold rotation $R_5 \in A_5$ between the symmetric mode v_0 and the electron mode v_1 ,

$$p_0 = |\langle v_0 | R_5 | v_1 \rangle|^2 = \frac{2}{9}.$$

From this one matrix element follow (Doc. 293, 367, 368):

1. **The Koide phase** $\theta = 2/9$ [B]: the same number, written as an angle in the Koide parametrisation, gives the three charged lepton masses without free parameters to 10^{-3} % (Doc. 292, 293, 367).
2. **The redistribution weights** $p_0 = 2/9$, $p_1 = (2 + 3\phi)/9$, $p_2 = (5 - 3\phi)/9$ with $p_0 + p_1 + p_2 = 1$ [B].
3. **The ϕ skeleton of the weight ratios** [B]:

$$\frac{p_1}{p_2} = \phi^8, \quad \frac{p_0}{p_2} = 2\phi^4.$$

Both relations are exact algebraic identities following from $p_1 = (2+3\phi)/9$ and $p_2 = (5-3\phi)/9$ (Doc. 368).

4. **The link $4/3 = 2Q$** (Doc. 371): the Koide quotient $Q = 2/3$ and the three-dimensionality factor $4/3$ of the electromagnetic mass (Abraham 1902) satisfy $4/3 = 2Q$ exactly.
The mechanism thus carries four independent structural results from a single matrix-element value $2/9$.

2 The Spectrum of the Matrix Elements [B]

Before asking which constant the mechanism might carry further, the reverse question must be answered: which numbers can it produce at all? This is a finite calculation. The group A_5 has 60 elements; in the three-dimensional representation, expressed in the $\mathbb{Z}_3$ mode basis v_0, v_1, v_2 , every element has nine matrix elements. The check script generates all 60 elements from R_5 and C_3 and collects the squared moduli. Exactly eight values occur:

$$\left\{ 0, \frac{5-3\phi}{9}, \frac{1}{9}, \frac{2}{9}, \frac{4}{9}, \frac{5}{9}, \frac{2+3\phi}{9}, 1 \right\} \\ = \{0, 0.0162, 0.1111, 0.2222, 0.4444, 0.5556, 0.7616, 1\}.$$

All denominators are $9 = |\mathbb{Z}_3|^2$; the two irrational values are the weights p_2, p_1 from Doc. 293/368. The set is not closed under $x \mapsto 1 - x$ ($1 - 1/9 = 8/9$ is missing), but the subset $\{4/9, 5/9\}$ and the pair $\{p_1, p_0 + p_2\}$ are.

Reach theorem. Every constant that arises as the squared modulus of a single A_5 matrix element in the $\mathbb{Z}_3$ basis lies in this eight-element set. What does not lie in it cannot be delivered by the mechanism in this form [B].

3 Test Against the Mixing Parameters of the Standard Model

The eight-element set is held against the dimensionless mixing parameters of the Standard Model (PDG 2024, NuFIT 6.0). Hits and misses are both recorded.

Weinberg angle: on-shell = $2/9$ [K]

The Weinberg angle has two common definitions. The $\overline{\text{MS}}$ value $\sin^2 \hat{\theta}_W(M_Z) = 0.2312$ depends on the renormalisation scheme. The on-shell definition

$$\sin^2 \theta_W \equiv 1 - \frac{M_W^2}{M_Z^2}$$

by contrast is fixed by pole masses. Since the FFGFT ground relation $\tilde{T} \cdot m = 1$ yields pole masses (Doc. 333), the on-shell scheme is the natural one for FFGFT.

Input	M_W [GeV]	$1 - M_W^2/M_Z^2$	to 2/9	Sign.
World average 2024 (without CDF)	80.3692	0.22321	0.44 %	3.8σ
CDF 2022	80.4335	0.22197	0.12 %	1.4σ
PDG on-shell direct	—	0.22305	0.37 %	3.6σ
MS (comparison)	—	0.23121	3.9 %	—

The value $2/9$ lies between the world average and the CDF value; the value of M_W that gives $\sin^2 \theta_W = 2/9$ exactly is

$$M_W(2/9) = M_Z \sqrt{7/9} = 80.420 \text{ GeV}.$$

This is a falsifiable prediction. It lies 3.8σ from the current world average and 1.4σ from the CDF value; the M_W measurements are at present not mutually consistent. The classification is **[K]** at 0.4 %, with the tension stated.

This replaces the previous corpus status: Doc. 336 named as the best Galois candidate $|\text{GF}(3)^*|/|\text{GF}(9)^*| = 1/4$ with 8 % deviation. The renormalisation-group running from ξ (Doc. 323, 0.2308) and the on-shell matrix element $2/9$ are two descriptions: the first in the $\overline{\text{MS}}$ scheme from the scale flow, the second in the on-shell scheme from the lattice geometry. Whether both describe the same geometric object remains open **[S]**; that they serve different schemes is settled.

Atmospheric mixing angle: $\sin^2 \theta_{23} \in \{4/9, 5/9\}$ **[K]**

The eight-element set contains the pair $4/9, 5/9$ with $4/9 + 5/9 = 1$. For the atmospheric neutrino mixing angle, $\sin^2 \theta_{23}$ is experimentally close to $1/2$, with the octant unresolved:

Fit	$\sin^2 \theta_{23}$	nearest candidate	significance
PDG 2024 (NO)	0.558 ± 0.020	$5/9 = 0.5556$	0.1σ
NuFIT 6.0 (IO)	0.550 ± 0.015	$5/9$	0.4σ
NuFIT 6.0 (NO)	0.470 ± 0.015	$4/9 = 0.4444$	1.7σ

In the matrix-element picture the octant ambiguity of the data is the ambiguity $4/9 \leftrightarrow 5/9$. The mechanism thus makes a prediction that is independent of the octant:

$$\left| \sin^2 \theta_{23} - \frac{1}{2} \right| = \frac{1}{18} = 0.0556.$$

Maximal mixing ($\sin^2 \theta_{23} = 1/2$) is excluded. This can be tested with DUNE and Hyper-Kamiokande. Classification **[K]** for the value, **[S]** for the assignment to the neutrino sector (the neutrino masses themselves come from $\text{GF}(27)^*$ in Doc. 340, not from A_5 ; that the angle θ_{23} is nevertheless an A_5 matrix element would be a second, independent A_5 access to the lepton sector).

Relation to the Galois reading in Doc. 340

Doc. 340 reads two mixing angles off the orbit elements of $\text{GF}(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$: $\sin^2 \theta_{12} \approx \cos^2(4\pi/13) = 0.323$ and $\sin^2 \theta_{23} \approx \cos^2(10\pi/13) = 0.560$. For θ_{23} two values therefore stand side by side, 0.560 and $5/9 = 0.556$, both of which hit the PDG value to within 0.4 %. This needs to be clarified.

The two values are not identical [B]. $\cos(2\pi/13)$ has a minimal polynomial of degree $\varphi(13)/2 = 6$ over $\mathbb{Q}$; consequently $\cos^2(10\pi/13)$ is irrational, whereas $5/9$ is rational. The closeness of 0.85 % is numerical, not algebraic: $5/9$ is merely the best rational approximation to $\cos^2(10\pi/13)$ with denominator ≤ 13 .

The Galois reading is an order-of-magnitude statement. The set $\{\cos^2(2\pi k/13), 1 - \cos^2(2\pi k/13) \mid k = 1..6\}$ contains twelve values between 0.01 and 0.99. Held against seven dimensionless parameters of the Standard Model, for six of them an element is found within 6.6 % — the set is too dense to discriminate at this tolerance. The eight-element set of §2, by contrast, discriminates: three hits below 1.3 %, the rest above 5 %. The $\cos^2$ readings in Doc. 340 are therefore to be understood as orders of magnitude, not as derivations; Doc. 340 itself labels them as “read off”.

Division of labour in the neutrino sector. This determines which structure carries which parameter:

Quantity	Structure	Source	Status
$\Delta m_{\text{atm}}^2, \Delta m_{\text{sol}}^2$	GF(27)* orbits	Doc. 340	[K] (1 %, 0.5 %)
ν_3 Dirac, $\nu_{1,2}$ Majorana	quadratic character mod 13	Doc. 343	[B]
$\sin^2 \theta_{13} = (25\xi)^{2/3}$	$\xi \cdot A_5 $ ($25\xi = 1/300$)	Doc. 340	[K] (1.4 %)
$\sin^2 \theta_{23} \in \{4/9, 5/9\}$	A_5 matrix element	Doc. 372	[K] (0.4 %)
$\sin^2 \theta_{12}$	only $\cos^2(4\pi/13)$, order of magnitude	Doc. 340	[S] (5 %)
δ_{CP}	orbit-3 element $k = 10$	Doc. 340	[S]

The neutrino sector is carried by two structures: GF(27)* for the masses and the Dirac/Majorana character, A_5 (alone or with ξ) for θ_{23} and θ_{13} . Both sit on the same lattice (Doc. 364: $A_5 \subset \text{Aut}(D_4)$ only via the phase channel GF(81); GF(27) as the ordering structure, Doc. 341). Whether θ_{12} is likewise an A_5 object that only becomes visible through a product or another basis, or whether it remains with $\mathbb{Z}_{13}$ alone, is the remaining open question of the sector **[S]**.

Cabibbo angle [S]

$\lambda = \sin \theta_C = 0.2250 \pm 0.0007$ lies 1.2 % (4.1σ) from $2/9$. The long-known numerical closeness $\sin \theta_C \approx \sin^2 \theta_W^{\text{on-shell}}$ is read here as the closeness of both quantities to $2/9$. This is not an identification: Doc. 041 derives $|V_{us}| = \sqrt{m_d/m_s} \cdot f_{\text{Cab}}$ from quark masses, and the quark coefficients are QCD-calibrated (Doc. 289). Whether λ is a matrix element of another subgroup of $\text{Aut}(D_4)$ is open **[S]**.

Negative findings [X]

Three parameters do not lie in the eight-element set and cannot be reached by a single A_5 matrix element:

Parameter	Value	nearest element	distance
$\sin^2 \theta_{12}$ (solar)	0.307	$2/9$	28 %
$\sin^2 \theta_{13}$ (reactor)	0.0222	$(5 - 3\phi)/9$	27 %
$\alpha_s(M_Z)$	0.118	$1/9$	5.8 %

The reach theorem of §2 makes this a sharp statement: these three quantities require either another group, a product or difference of matrix elements, or a different mechanism.

For α_s the corpus has the Galois running with ξ ; for θ_{12} and θ_{13} nothing is available in the matrix-element picture.

Quarks: compact description present, unrolling incompatible

In the corpus the charged leptons have two descriptions that meet: compact, the Yukawa ladder $m = r\xi^p v$ with $r_e = 4/3$ (Doc. 006), and unrolled ($T^4 \rightarrow \mathbb{R}^3 \times S^1$, Doc. 370), the A_5 mechanism of this document. Both arrive at $Q = 2/3$, $r = \sqrt{2}$ — the “special point” of Doc. 289.

The quarks have only the compact description, and it is more developed than the comparison with unrolling suggests. The exponents p_i are derived from the torus topology (Doc. 189: $p = d_{\text{akt}}/3$, one active dimension fewer per step) — up/down $3/2$, strange 1, charm $2/3$, bottom $1/2$ as transition exponent, top $-1/3$ as inverse winding **[K]**. The ξ scale correction and the cumulative factor $K_{\text{frak}} = 1 - 100\xi$ from the scale flow Planck $\rightarrow$ electroweak (Doc. 133) weight the ladder; the extended fractal method thereby reaches $\sim 1.2\%$ without free parameters (Doc. 005). What is missing are the prefactors r_i : Doc. 189 sets $r_i = n_\phi^2/(n_\theta n_c)$ with $n_c = 3$ for quarks and explicitly names the complete derivation as outstanding; the r_i tabulated in Doc. 006 are back-calculated from the measured masses, but carry exclusively prime factors from the Galois grid $\{2, 3, 5, 7, 11, 13\}$ **[B]** (Doc. 358, Theorem B). The gap is therefore precise: p_i derived, r_i postulated **[S]**.

The unrolling cannot be transferred, and Doc. 289 shows why: the quark triplets do not sit at the special point. With PDG-like values, $Q_{\text{up}} = 0.849$ ($r = 1.76$) and $Q_{\text{down}} = 0.730$ ($r = 1.54$). Scale dependence does not change this: running all six masses to a common scale ($\overline{MS}$ at M_Z , m_t or roughly M_{GUT}), Q_{up} stays between 0.85 and 0.89 and Q_{down} between 0.72 and 0.75 — never close to $2/3$ **[K]**. The A_5 mechanism, however, produces *exactly* $Q = 2/3$ — that is its signature ($\theta = |A|^2 = 2/9 \Rightarrow Q = 2/3$, Doc. 367), and the eight-element set contains no value that yields $Q = 0.85$ or 0.73 . The quark masses are incompatible with the lepton unrolling mechanism **[X]**, not merely unworked.

Three structural reasons, each documented in the corpus: quarks carry colour, and the $\mathbb{Z}_3$ that counts the generations for leptons is at the same time the colour $\mathbb{Z}_3$ for quarks (Doc. 321) — the mode space would be 3×3 , not 3. Quarks come as isospin doublets, six instead of three modes per sector. And $\tilde{T} \cdot m = 1$ gives pole masses, which are not defined for u, d, s (Doc. 333). The open question is therefore not “how does one unroll quarks like leptons” but: which other subgroup of $\text{Aut}(D_4)$ or which tensor product $A_5 \otimes \mathbb{Z}_3^{\text{colour}}$ has a matrix-element spectrum that contains $Q \approx 0.85$ and 0.73 . Doc. 358 names the field $\text{GF}(729)$ as the entry point for the top coefficient (prime 7, $k = 6$) **[S]**. That is a different programme, not an appendix to lepton unrolling.

Gauge bosons and Higgs

What is derived. The electroweak scale v follows by two routes to 0.3% **[K]**: via the ladder with m_e as anchor, $m_e/((4/3)\xi^{3/2}) = 248.9$ GeV without and $248.9 \cdot 74/75 = 245.6$ GeV (-0.25%) with K_{frak} (Docs. 006, 373, 375), or via $v/E_P = \xi^4/(5\pi)$, whose factor 10 is fitted to the measured value (Doc. 387). Both routes thus contain a measured input. The ratio M_W/M_Z follows from §3: $M_W^2/M_Z^2 = 1 - 2/9 = 7/9$ **[K]**. The fine-structure constant $1/\alpha = 3700/27$ follows from Galois orders (Doc. 338) **[K]**. Thus v , α and $\sin^2 \theta_W$ are available; the absolute masses $M_W = gv/2$, $M_Z = M_W/\cos \theta_W$ (Doc. 320) follow from them via the standard electroweak relations up to radiative corrections (Δr , of the order of a per cent), which the corpus does not compute. Photon and the eight gluons are massless: through the unbroken $U(1)_{\text{EM}}$ and $SU(3)_c$ (Doc. 320) **[B]**, where $SU(3)_c$ follows from $\mathbb{Z}_3$ triality (Doc. 321) and the eight gluons from the eight three-element orbits of $\text{GF}(27)^*$ (Doc. 339) **[B]**.

Where the Higgs stands on the ladder. The p ladder of Doc. 189 places the Higgs scale at $p = 0$ ($d_{\text{akt}} = 0$): $m_h = r_h v$ without ξ suppression, with $r_h = m_h/v = 0.508$. This correctly places the Higgs as the only particle on the v scale but provides no prefactor — the same gap as for the quark r_i .

What is not derived: the Higgs mass [X]. The corpus contains two formulas for m_h , and neither survives recalculation. Doc. 041 gives $m_h = v\xi^{1/4}$ with the numerical value 125 GeV; in fact $v\xi^{1/4} = 26.5$ GeV. The same table gives $\Lambda_{\text{QCD}} = v\xi^{1/3} = 200$ MeV; in fact 12.6 GeV. Doc. 005 gives $m_H = m_t \phi(1 + \xi D_f) = 124.95$ GeV; since $m_t \phi = 279.4$ GeV, this requires $\xi D_f = -0.553$, i.e. $D_f \approx -4100$ — a fit parameter, not a derivation. These two formulas are recorded as R122 in Doc. 190. Viable candidates are only provided by Doc. 373 (Galois chain and trace rule). The eight-element set yields nothing either: $m_h/v = 0.508$ and $m_h^2/v^2 = 0.258$ do not lie in it [X].

Systematic search. Instead of guessing individual formulas, 17 dimensionless ratios of m_h, M_W, M_Z, m_t, v were held against a library of 276 FFGT-related numbers (fractions with denominator ≤ 12 , powers of ϕ , roots, Galois orders, K_{frak} , the eight-element set). At 0.5 % tolerance twelve hits are found, at 0.2 % three; the random expectation is about fourteen and six respectively. There is therefore *no* signal above the look-elsewhere background [X]. The obvious reading $m_h = v/2$ lies at 1.7 % and is not distinguished.

A single candidate has internal structure and is therefore recorded, explicitly as selected after the fact: $m_h/M_Z = 1.3730 \pm 0.0012$ and $m_t/m_h = 1.3784 \pm 0.0026$ both lie at $11/8 = 1.375$ (at 1.7σ and 1.3σ respectively), hence

$$M_Z : m_h : m_t \approx 1 : \frac{11}{8} : \left(\frac{11}{8}\right)^2, \quad m_h \approx \sqrt{m_t M_Z} \quad (0.20\%, 1.6\sigma).$$

The numbers $11 = \max(\text{orbit}_4)$ and $8 = |\text{GF}(9)^*|$ are both Galois quantities of the corpus (Doc. 340, 336). The Higgs would then sit as the geometric mean between Z and top — in the language of $\tilde{T} \cdot m = 1$: its winding number is the mean of the two. Status [S], with the proviso that one candidate out of about 5000 tests is not evidence but a hypothesis that would need an independent derivation.

4 Hadrons: Bound States on the Lattice

The leptons are unrolled primary modes of the T^4/D_4 lattice — free, asymptotic particles whose masses follow directly from $\tilde{T} \cdot m = 1$. Hadrons are not. There is no free quark; what is measured is the system energy of the bound state. Three consequences can be tested numerically.

Koide quotient of the hadron multiplets [B]. For an isospin multiplet of dimension n with nearly degenerate masses $m_i = m(1 + \delta_i)$, $|\delta_i| \ll 1$, it holds algebraically exactly that

$$Q \cdot n = 1, \quad Q = \frac{\sum m_i}{(\sum \sqrt{m_i})^2}.$$

Numerically: $Q(p, n) = 1/2$, $Q(\Sigma^+, \Sigma^0, \Sigma^-) = 1/3$, $Q(\Delta^{++}, \Delta^+, \Delta^0, \Delta^-) = 1/4$, $Q(\Omega^-) = 1$, all to $< 0.01\%$ [B]. This is an algebraic identity for degenerate masses, not a new result — but it sharpens the contrast: the leptons break exactly this relation. For $n = 3$ generations the degeneracy rule would require $Q_{\text{Lep}} = 1/3$; in fact $Q_{\text{Lep}} = 2/3$. The factor 2 is the signature of the A_5 mechanism: the icosahedron lifts the degeneracy specifically such that $Q = 2 \cdot (1/3) = 2/3$. Hadrons remain at the degenerate relation because their $SU(2)$ isospin symmetry is almost exactly conserved.

Gell-Mann–Okubo splitting consistent with m_s [K]. The baryon decuplet masses show a uniform spacing: $\Delta m(\Sigma^* - \Delta) = 151.7$ MeV, $\Delta m(\Xi^* - \Sigma^*) = 148.1$ MeV, $\Delta m(\Omega - \Xi^*) = 140.4$ MeV, mean 146.7 MeV. This is the Gell-Mann–Okubo rule; the splitting corresponds to the effective strange quark mass. From the Galois coefficient r_s (Doc. 006) follows $m_s^{(\text{FFGFT})} = r_s \xi v \approx 93$ MeV. The agreement is consistent to a factor of 1.6 [K], but not a precision statement — the difference lies in the QCD dynamics, not in the lattice geometry.

Λ_{QCD} not derivable from the ξ ladder [X]. $\Lambda_{\text{QCD}} \approx 210$ MeV (PDG, $n_f = 5$) could be written as $v \xi^p$ for $p \approx 0.79$ — not a step of the ladder $p = n/3$ or $n/2$. The route via the hadron radius ($\omega_{\min} = 1/R_{\text{had}}$, Doc. 319) is also circular as long as R does not follow from ξ . Λ_{QCD} is thus the sharpest open question of the hadron sector: it lies neither in the eight-element set (§2), nor on the ξ ladder, nor in the Galois grid of Doc. 358.

Gluons as standing waves. Doc. 319 describes gluons as standing waves in the closed proton torus: massless, pure field energy ($E = pc$, $d\tau = 0$), topologically confined by the $\mathbb{Z}_3/SU(3)$ boundary condition (eight linking states of the three flux strands = eight generators of $SU(3)_c$, Doc. 321 [B]). In this picture confinement is not a dynamical mechanism but a topological impossibility of escape: the closed torus admits only integer wave numbers; there is no free propagation to the outside. In this language Λ_{QCD} is the minimum frequency on the torus, $\omega_{\min} = 1/R_{\text{Proton}}$ (Doc. 319 [S]).

R_{Proton} structurally classified [S]. Doc. 319 gives $\Lambda_{\text{QCD}} \approx \omega_{\min} = \hbar c / R_{\text{Proton}}$ [S]. The FFGFT quantities allow a structural classification: the geometric mean of the classical electron radius $r_e = \alpha \hbar c / m_e$ and the proton Compton wavelength $\lambda_p = \hbar c / m_p$ is

$$R_{\text{geom}} = \sqrt{r_e \lambda_p} = \frac{\hbar c \sqrt{\alpha}}{\sqrt{m_e m_p}} = 0.770 \text{ fm},$$

8.5 % below the measured charge radius 0.841 fm. All three factors — α from Galois structure (Doc. 338), m_e from the A_5 mechanism, m_p from the ξ ladder (Doc. 189) — are available in FFGFT. The missing factor $R_{\text{PDG}}/R_{\text{geom}} \approx 1.09$ lies in the quark–gluon binding dynamics, which the programme does not yet capture formally.

Conclusion. Hadrons are bound states on the same lattice as leptons — the same ξ , the same ladder for p_i , the same Galois grid for r_i . Confinement is a topological boundary condition, not a separate mechanism. R_{Proton} is structurally classified to 8.5 %; the residual lies in the binding dynamics. This is the honest limit [S].

5 The More General Principle

The four results of §1 suggest a more general thesis:

The matrix-element principle: every dimensionless physical constant that arises from the overlap of two lattice modes under a finite group rotation can be read as the squared modulus of a matrix element $|\langle v_i | g | v_j \rangle|^2$ — where g is a group element, v_i, v_j are orthonormal modes of the T^4/D_4 lattice, and the result has no free parameters.

For the lepton masses this principle is fully carried out with $g = R_5 \in A_5, i = 0, j = 1$ **[B]**. For other sectors it is an open hypothesis **[S]**. The sharpest test question is: is there a finite subgroup of $\text{Aut}(D_4)$ and two lattice modes whose squared matrix-element modulus reproduces the Weinberg angle or the Cabibbo angle without free parameters?

That $|A_5| = 60 = 4 \cdot 3 \cdot 5$ unites the three defining symmetry numbers of the lepton structure, and that A_5 is the only simple finite subgroup of $SO(3)$ with five-fold elements, make this hypothesis structurally attractive.

6 Result

1. **Reach theorem [B]**: in the $\mathbb{Z}_3$ basis A_5 generates exactly eight squared matrix-element moduli $\{0, (5 - 3\phi)/9, 1/9, 2/9, 4/9, 5/9, (2 + 3\phi)/9, 1\}$.
2. **Lepton masses**: from $2/9$ follow the Koide phase, the weights, the ϕ skeleton ($p_1/p_2 = \phi^8$, $p_0/p_2 = 2\phi^4$) and $4/3 = 2Q$ **[B]**.
3. **Weinberg angle [K]**: on-shell $1 - M_W^2/M_Z^2 = 2/9$ to 0.4% ; prediction $M_W = M_Z\sqrt{7/9} = 80.420 \text{ GeV}$, between the world average and CDF. Replaces the candidate $1/4$ (8 %) of Doc. 336. The $\overline{\text{MS}}$ result 0.2308 of Doc. 323 remains as the scale-flow description.
4. **Atmospheric mixing [K]**: $\sin^2 \theta_{23} \in \{4/9, 5/9\}$, prediction $|\sin^2 \theta_{23} - 1/2| = 1/18$; maximal mixing excluded. The PDG value 0.558 hits $5/9$ at 0.1σ .
5. **Comparison with Doc. 340 [B]**: $\cos^2(10\pi/13) \neq 5/9$ (degree 6 versus rational); the $\cos^2(2\pi k/13)$ readings are orders of magnitude. Division of labour: $\text{GF}(27)^*$ carries masses and Dirac/Majorana, A_5 carries θ_{23} and (with ξ) θ_{13} ; θ_{12} remains open **[S]**.
6. **Quarks [X]**: compact ladder with derived p_i (Doc. 189) and ξ/K_{frak} weighting (Doc. 133, 005), $\sim 1.2\%$ without free parameters; r_i postulated, in the Galois grid **[B]**. $Q_{\text{up}} \approx 0.85\text{--}0.89$, $Q_{\text{down}} \approx 0.72\text{--}0.75$ at all scales: incompatible with A_5 unrolling ($Q = 2/3$ exactly).
7. **Bosons**: $\nu, \alpha, \sin^2 \theta_W$ available **[K]**; photon/gluons massless **[B]**; the Higgs formulas in Doc. 041 and 005 do not survive recalculation **[X]** (R122); candidates in Doc. 373.
8. **Cabibbo [S]**: λ within 1.2% of $2/9$, not identified; Doc. 041 has a different derivation.
9. **Negative [X]**: $\sin^2 \theta_{12}, \sin^2 \theta_{13}, \alpha_s$ do not lie in the eight-element set.
10. The general matrix-element principle (§5) thus has three confirmed instances ($2/9$ lepton, $2/9$ Weinberg, $\{4/9, 5/9\}$ atmospheric) and three excluded ones; the next question is whether another subgroup of $\text{Aut}(D_4)$ carries the excluded values.

Check script:

2/python/Dok372_Skripte/pruef_372_matrixelement.py

Register (Doc. 190): R122 (Higgs mass in Doc. 041 and 005).

Bibliography

- [1] J. Pascher, *Doc. 292 — Koide formula and FFGFT* (2025).
- [2] J. Pascher, *Doc. 293 — Icosahedral matrix element* (2025).
- [3] J. Pascher, *Doc. 323 — Renormalisation group and Weinberg angle* (2025).
- [4] J. Pascher, *Doc. 340 — Neutrino mass hierarchy from $GF(27)^*$* (2026).
- [5] J. Pascher, *Doc. 367 — Why $\theta = 2/9$* (2026).
- [6] J. Pascher, *Doc. 368 — From p_0, p_1, p_2 to lepton masses* (2026).
- [7] J. Pascher, *Doc. 371 — The factor $4/3$* (2026).